Systems of linear and quadratic equations

ANSWER KEY

Core Concepts: Systems of linear and quadratic equations

1 For the model $f(x) = 3x + 3$ in systems of linear and quadratic equations, evaluate $f(2)$.

a. 9 ✓

b. 12

c. 6

d. 13

Substitute $x = 2$: $f(2) = 3(2) + 3 = 9$.

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

3 A data set recorded for systems of linear and quadratic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 3x + 3$ in systems of linear and quadratic equations, evaluate $f(3)$.

a. 12 ✓

b. 15

c. 9

d. 16

Substitute $x = 3$: $f(3) = 3(3) + 3 = 12$.

5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

6 A data set recorded for systems of linear and quadratic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

Graphical & Geometric Interpretation

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7 For the model $f(x) = 3x + 3$ in systems of linear and quadratic equations, evaluate $f(4)$.

a. 15 ✓

b. 18

c. 12

d. 19

Substitute $x = 4$: $f(4) = 3(4) + 3 = 15$.

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

9 A data set recorded for systems of linear and quadratic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 3x + 3$ in systems of linear and quadratic equations, evaluate $f(5)$.

a. 18 ✓

b. 21

c. 15

d. 22

Substitute $x = 5$: $f(5) = 3(5) + 3 = 18$.

11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ **and** $x = 9$ ✓

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

12 A data set recorded for systems of linear and quadratic equations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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